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VEHICLE BRAKE SYSTEM AND VEHICLE HAVING SAME

Abstract

A vehicle brake system includes: a plurality of wheel cylinders, a main pressure building module, a redundant pressure building module and a brake master cylinder assembly; the main pressure building module is separately connected to the plurality of wheel cylinders and is used for conveying a brake fluid to the plurality of wheel cylinders; the redundant pressure building module is connected to at least one wheel cylinder and is used for conveying the brake fluid to the at least one wheel cylinder; and the brake master cylinder assembly is separately connected to the plurality of wheel cylinders and is used for conveying the brake fluid to the plurality of wheel cylinders. Further disclosed is a vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present disclosure is a bypass continuation of PCT International Application No. PCT/CN2023/107884, filed on Jul. 18, 2023, which claims priority to Chinese Patent Application No. 202211514374.6, filed on Nov. 30, 2022 and entitled “VEHICLE BRAKE SYSTEM AND VEHICLE HAVING SAME”. The entire contents of the above-referenced applications are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the field of vehicle technologies, and more specifically, a vehicle brake system and a vehicle having same.

BACKGROUND

[0003] A vehicle brake system in a related system usually includes multiple wheel cylinders, a pressure building module, and a master brake cylinder assembly. The pressure building module is configured to convey a brake fluid to the multiple wheel cylinders, to output a brake pressure to brake a vehicle. However, when the pressure building module has a fault and cannot output a braking force, or the brake pressure outputted by the pressure building module cannot meet a braking demand, the vehicle has an insufficient braking force, resulting in low braking safety.

SUMMARY

[0004] The present disclosure is intended to resolve at least one of technical problems in the related art. One objective of the present disclosure is to provide a vehicle brake system. The vehicle brake system has advantages such as strong braking force output, a high braking response speed, and high braking safety.

[0005] Embodiment of the present disclosure further provides a vehicle having the foregoing vehicle brake system.

[0006] To achieve the foregoing objective, an embodiment of a first aspect of the present disclosure provides a vehicle brake system, which includes multiple wheel cylinders, a master pressure building module, a redundant pressure building module, and a master brake cylinder assembly. The master pressure building module is respectively connected with the multiple wheel cylinders, and is configured to convey a brake fluid to the multiple wheel cylinders. The redundant pressure building module is connected with at least one wheel cylinder, and is configured to convey a brake fluid to the at least one wheel cylinder. The master brake cylinder assembly is respectively connected with the multiple wheel cylinders, and is configured to convey a brake fluid to the multiple wheel cylinders.

[0007] The vehicle brake system in this embodiment of the present disclosure is provided not only with the master pressure building module and the master brake cylinder assembly, but also with the redundant pressure building module, which resolves a problem of an insufficient braking force of the vehicle and relatively low braking safety as a result of a brake pressure failing to meet a braking demand when the pressure building module of the vehicle brake system is faulty and cannot output a braking force, which has advantages such as strong braking force output, a high braking response

speed, and high braking safety.

[0008] According to some embodiments of the present disclosure, the vehicle brake system has multiple braking modes between which the vehicle brake system is automatically switchable. In different braking modes, the master pressure building module, the redundant pressure building module, or the master brake cylinder assembly separately convey a brake fluid to the wheel cylinders, or the master pressure building module and the redundant pressure building module jointly convey a brake fluid to the wheel cylinders, or the master pressure building module and the master brake cylinder assembly jointly convey a brake fluid to the wheel cylinders, or the redundant pressure building module and the master brake cylinder assembly jointly conveying a brake fluid to the wheel cylinders.

[0009] According to some embodiments of the present disclosure, the master brake cylinder assembly and the master pressure building module are arranged separately and spaced away.

[0010] According to some embodiments of the present disclosure, the master brake cylinder assembly is mounted to the redundant pressure building module.

[0011] According to some embodiments of the present disclosure, a first output flow channel is constructed in the master pressure building module. The first output flow channel is connected with the multiple wheel cylinders. A second output flow channel is constructed in the redundant pressure building module. The second output flow channel is connected with the at least one wheel cylinder.

[0012] According to some embodiments of the present disclosure, the vehicle brake system further includes a first switching flow channel. The first switching flow channel is respectively connected with the first output flow channel and the master brake cylinder assembly. The master brake cylinder assembly is connected with the multiple wheel cylinders through the first switching flow channel and the first output flow channel.

[0013] According to some embodiments of the present disclosure, the vehicle brake system further includes a second switching flow channel. The second switching flow channel is respectively connected with the second output flow channel and the master brake cylinder assembly. The master brake cylinder assembly is connected with the multiple wheel cylinders through the second switching flow channel and the second output flow channel.

[0014] According to some embodiments of the present disclosure, the vehicle brake system further includes a master control valve, a redundant control valve, and a master cylinder control valve. The master control valve is arranged in the first output flow channel, and is configured to control connection/disconnection between the master pressure building module and the multiple wheel cylinders. The redundant control valve is arranged in the second output flow channel, and is configured to control connection/disconnection between the redundant pressure building module and the at least one wheel cylinder. The master cylinder control valve is mounted to at least one of the master pressure building module and the redundant pressure building module. The master cylinder control valve is connected between the master brake cylinder assembly and the multiple wheel cylinders, and is configured to control connection/disconnection between the master brake cylinder assembly and the multiple wheel cylinders.

[0015] According to some embodiments of the present disclosure, the vehicle brake system further includes a brake pedal, a displacement sensor, a wheel cylinder pressure sensor, and a controller. The brake pedal is connected with the master brake cylinder assembly. The displacement sensor is configured to detect a movement position of at least one of the brake pedal and a piston of the master brake cylinder assembly. The wheel cylinder pressure sensor is connected with the multiple wheel cylinders, and is configured to detect pressures of the multiple wheel cylinders. The controller is respectively connected with the displacement sensor, a vehicle wheel speed sensor, the wheel cylinder pressure sensor, the master control valve, the redundant control valve, and the master cylinder control valve. The controller is configured to control closing/opening of the master control valve, the redundant control valve, and the master cylinder control valve based on a detection result of the displacement sensor and a detection result of at least one of the vehicle

wheel speed sensor and the wheel cylinder pressure sensor.

[0016] According to some embodiments of the present disclosure, the controller is further respectively connected with the master pressure building module and the redundant pressure building module. The controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, the master pressure building module and the redundant pressure building module whether to build a pressure.

[0017] According to some embodiments of the present disclosure, multiple controllers arranged. The multiple controllers include a master module controller and a redundant module controller. The master module controller is respectively connected with the displacement sensor, the vehicle wheel speed sensor, the wheel cylinder pressure sensor, the master control valve, the redundant control valve, and the master cylinder control valve. The master module controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, closing/opening of the master control valve and the master pressure building module whether to build a pressure. The redundant module controller is respectively connected with the displacement sensor, the vehicle wheel speed sensor, the wheel cylinder pressure sensor, the master control valve, the redundant control valve, and the master cylinder control valve. The redundant module controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, closing/opening of the redundant control valve and whether the redundant pressure building module to build a pressure. The master module controller or the redundant module controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, the closing/opening of the master cylinder control valve.

[0018] According to some embodiments of the present disclosure, the vehicle brake system further includes a first angle sensor, a first pressure sensor, a second angle sensor, a second pressure sensor, and a third pressure sensor. The first angle sensor is configured to detect an angle change of a motor of the master pressure building module. The first pressure sensor is configured to detect a pressure of a brake fluid conveyed by the master pressure building module. The second angle sensor is configured to detect an angle change of a motor of the redundant pressure building module. The second pressure sensor is configured to detect a pressure of a brake fluid conveyed by the redundant pressure building module. The third pressure sensor is configured to detect a pressure of conveyed brake fluid of the master brake cylinder assembly. The controller is further respectively with the first angle sensor, the first pressure sensor, the second angle sensor, the second pressure sensor, and the third pressure sensor. The controller is configured to: determine whether the master pressure building module is faulty based on detection results of the first angle sensor and the first pressure sensor, determine whether the master pressure building module is faulty based on detection results of the second angle sensor and the second pressure sensor, and determine whether the master brake cylinder assembly is faulty based on a detection result of the third pressure sensor.

[0019] According to some embodiments of the present disclosure, the vehicle brake system further includes a master liquid reservoir and a redundant liquid reservoir. The master liquid reservoir is connected with the master pressure building module. The redundant liquid reservoir is respectively connected with the redundant pressure building module, the master brake cylinder assembly, and the master liquid reservoir.

[0020] According to some embodiments of the present disclosure, the vehicle brake system further includes a brake pedal, a stroke simulator, and a simulator switch valve. The brake pedal is connected with the master brake cylinder assembly. The stroke simulator is respectively connected with the master brake cylinder assembly and the brake pedal, and is configured to feed back a

damping force to the brake pedal. The simulator switch valve is separately connected with the stroke simulator and the master brake cylinder assembly, and is configured to control on/off of the stroke simulator and the master brake cylinder assembly.

[0021] According to some embodiments of the present disclosure, the vehicle brake system further includes multiple supercharging valves and multiple pressure reducing valves. The multiple pressure increasing valves are connected with the multiple wheel cylinders in one-to-one correspondence. Each pressure increasing valve is respectively connected with the master pressure building module, the redundant pressure building module, and the master brake cylinder assembly. The multiple pressure reducing valves are connected with the multiple wheel cylinders in one-to-one correspondence. Each pressure reducing valve is connected with the liquid reservoir.

[0022] According to an embodiment of a second aspect of the present disclosure, a vehicle is proposed, which includes the vehicle brake system according to the embodiment of the first aspect of the present disclosure.

[0023] The vehicle in the embodiment of the second aspect of the present disclosure uses the vehicle brake system in the embodiment of the first aspect of the present disclosure, which has advantages such as strong braking force output, a high braking response speed, and high braking safety.

[0024] Additional aspects and advantages of the present disclosure are to be partially given in the following description, and partially become apparent in the following description or learned through the practice of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The foregoing and/or additional aspects and advantages of the present disclosure become apparent and comprehensible from description of embodiments made with reference to the following drawings.

[0026] FIG. 1 is a principle diagram of a vehicle brake system according to an embodiment of the present disclosure;

[0027] FIG. 2 is a schematic block diagram of a vehicle brake system according to an embodiment of the present disclosure;

[0028] FIG. 3 is a schematic block diagram of a controller shown in FIG. 2; and

[0029] FIG. 4 is a schematic block diagram of a vehicle according to an embodiment of the present disclosure.

[0030] In the drawings: [0031] Vehicle brake system 1, [0032] Wheel cylinder 100, [0033] Master pressure building module 200, First output flow channel 210, Master control valve 220, First angle sensor 230, First pressure sensor 240, Master liquid reservoir 250, [0034] Redundant pressure building module 300, Second output flow channel 310, Redundant control valve 320, Second angle sensor 330, Second pressure sensor 340, Redundant liquid reservoir 350, [0035] Master brake cylinder assembly 400, First switching flow channel 410, Second switching flow channel 420, Master cylinder control valve 430, Third pressure sensor 440, Piston 450 [0036] Brake pedal 500, Displacement sensor 510, [0037] Stroke simulator 600, Simulator switch valve 610, [0038] Pressure increasing valve 700, Pressure reducing valve 800.

DETAILED DESCRIPTION

[0039] Embodiments described with reference to the drawings are exemplary, and the embodiments of the present disclosure are described in detail below.

[0040] In the description of the present disclosure, it should be understood that orientation or position relationships indicated by the terms such as “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “up”, “down”, “front”, “rear”, “left”, “right”, “vertical”,

“horizontal”, “top”, “bottom”, “inside”, “outside”, “clockwise”, “anticlockwise”, “axial direction”, “radial direction”, and “circumferential direction” are based on orientation or position relationships shown in the drawings, and are merely used for ease and brevity of description of the present disclosure, rather than indicating or implying that the mentioned device or element needs to have a particular orientation or be constructed and operated in a particular orientation. Therefore, such terms should not be construed as a limitation on the present disclosure.

[0041] In the description of the present disclosure, a “first feature” and a “second feature” may include one or more features.

[0042] In the description of the present disclosure, “multiple” means two or more.

[0043] In the description of the present disclosure, that the first feature is “on” or “under” the second feature may include that the first feature and the second feature are in direct contact, or may include that the first feature and the second feature are not in direct contact but are in contact by using other features therebetween.

[0044] In the description of the present disclosure, that the first feature is “on”, “above”, or “over” the second feature includes that the first feature is directly above and obliquely above the second feature, or merely indicates that the first feature is at a higher horizontal position than the second feature.

[0045] A vehicle brake system **1** in an embodiment of the present disclosure is described below with reference to the drawings.

[0046] As shown in FIG. **1**, the vehicle brake system **1** in this embodiment of the present disclosure includes multiple wheel cylinders **100**, a master pressure building module **200**, a redundant pressure building module **300**, and a master brake cylinder assembly **400**.

[0047] The master pressure building module **200** is respectively connected with the multiple wheel cylinders **100**, and is configured to convey a brake fluid to the multiple wheel cylinders **100**. The redundant pressure building module **300** is connected with at least one wheel cylinder **100**, and is configured to convey a brake fluid to the at least one wheel cylinder **100**. The master brake cylinder assembly **400** is respectively connected with the multiple wheel cylinders **100**, and is configured to convey a brake fluid to the multiple wheel cylinders **100**.

[0048] In the vehicle brake system **1**, the master pressure building module **200** may mainly output a brake pressure, and the redundant pressure building module **300** may assist in outputting a brake pressure. In other words, when the master pressure building module **200** fails or the brake pressure of the master pressure building module **200** is insufficient, the redundant pressure building module **300** may output a brake pressure to brake a vehicle. In addition, when the master pressure building module **200** and the redundant pressure building module **300** are both faulty, the vehicle may be braked directly through the master brake cylinder assembly **400**. The vehicle brake system **1** is provided not only with the master pressure building module **200** and the master brake cylinder assembly **400**, but also with the redundant pressure building module **300**, which resolves a problem of an insufficient braking force of the vehicle and relatively low braking safety as a result of a brake pressure failing to meet a braking demand when the pressure building module of the vehicle brake system **1** is faulty and cannot output a braking force.

[0049] For example, the redundant pressure building module **300** may be directly connected with the wheel cylinder **100**, or the redundant pressure building module **300** is connected with the wheel cylinder **100** through the master pressure building module **200**. In other words, the redundant pressure building module **300** is indirectly connected with the wheel cylinder **100**.

[0050] According to the vehicle brake system **1** in this embodiment of the present disclosure, the master pressure building module **200** is respectively connected with the multiple wheel cylinders **100**, and is configured to convey the brake fluid to the multiple wheel cylinders **100**. Therefore, when the vehicle needs to be braked, the master pressure building module **200** can simultaneously convey the brake fluid to the multiple wheel cylinders **100**, to implement pressure building of the wheel cylinders **100**. Further, the brake pressure can be outputted by the master pressure building

module **200** to brake the vehicle.

[0051] In addition, the redundant pressure building module **300** is connected with at least one wheel cylinder **100**, and is configured to convey the brake fluid to the at least one wheel cylinder **100**. In other words, the redundant pressure building module **300** may convey the brake fluid to only one wheel cylinder **100** or a part of the multiple wheel cylinders **100**. When the master pressure building module **200** is faulty or the brake pressure of the master pressure building module **200** is insufficient, the redundant pressure building module **300** may convey the brake fluid to a part of the multiple wheel cylinders **100**, so that the redundant pressure building module **300** may assist in braking of the vehicle through a part of the wheel cylinders **100**, thereby improving the braking safety of the vehicle brake system **1**. In addition, the redundant pressure building module **300** does not need to be in communication with the wheel cylinders **100**, which helps simplify a structure of the vehicle brake system **1** and facilitate an arrangement.

[0052] In some embodiments of the present disclosure, the redundant pressure building module **300** may be in communication with the multiple wheel cylinders **100**. The redundant pressure building module **300** may simultaneously convey the brake fluid to the wheel cylinders **100**, or may selectively convey the brake fluid to a part of the multiple wheel cylinders **100** based on a braking condition. When the master pressure building module **200** is faulty or a brake pressure of the master pressure building module **200** is insufficient, the redundant pressure building module **300** may simultaneously convey the brake fluid to the multiple wheel cylinders **100**, so that the redundant pressure building module **300** may assist in braking of the vehicle through the wheel cylinders **100**, thereby improving the braking safety of the vehicle brake system **1**. In addition, the redundant pressure building module **300** is in communication with the wheel cylinders **100**, the redundant pressure building module **300** can simultaneously convey the brake fluid to the multiple wheel cylinders **100**. Therefore, the redundant pressure building module **300** can simultaneously build a pressure for the multiple wheel cylinders **100**, the brake pressure outputted by the redundant pressure building module **300** is more balanced, and the braking effect is better.

[0053] In this way, the redundant pressure building module **300** can help the master pressure building module **200** to build a pressure for the multiple wheel cylinders **100**. When the master pressure building module **200** is faulty, the redundant pressure building module **300** may output the brake pressure, so as to ensure that the vehicle brake system **1** can normally brake, which helps improve the braking safety of the vehicle. When the braking force outputted by the master pressure building module **200** cannot meet the braking demand, the redundant pressure building module **300** and the master pressure building module **200** may output the braking force together, so as to improve the braking force outputted by the vehicle brake system **1**. As a result, the braking force outputted by the vehicle brake system **1** is stronger, and the braking effect is better. In addition, the master pressure building module **200** and the redundant pressure building module **300** simultaneously respond to the output braking force, so that a braking response speed is higher.

[0054] In addition, the master brake cylinder assembly **400** is respectively connected with the multiple wheel cylinders **100**, and is configured to convey the brake fluid to the multiple wheel cylinders **100**. In this way, when the master pressure building module **200** and the redundant pressure building module **300** are both faulty and cannot normally output the braking force, the vehicle brake system **1** may directly use the master brake cylinder assembly **400** to directly convey the brake fluid to the multiple wheel cylinders **100**, so that the master brake cylinder assembly **400** can directly build the pressure for the wheel cylinder **100** to brake the vehicle. After the braking of the vehicle is completed, fault information of the master pressure building module **200** and the redundant pressure building module **300** is checked, which further improves the braking safety of the vehicle.

[0055] As a result, the vehicle brake system **1** in this embodiment of the present disclosure has advantages such as strong braking force output, a high braking response speed, and high braking safety.

[0056] In some specific embodiments of the present disclosure, the vehicle brake system **1** has multiple braking modes between which the vehicle brake system is automatically switchable. In different braking modes, the master pressure building module **200**, the redundant pressure building module **300**, or the master brake cylinder assembly **400** separately conveys the brake fluid to the wheel cylinder **100**.

[0057] In other words, the vehicle brake system **1** has a mode in which the master pressure building module **200** brakes alone. In this case, the master pressure building module **200** conveys the brake fluid to the wheel cylinder **100** alone. In this case, the master pressure building module **200** brakes the vehicle alone.

[0058] The vehicle brake system **1** has a mode in which the redundant pressure building module **300** brakes alone. In this case, the redundant pressure building module **300** conveys the brake fluid to the wheel cylinder **100** alone. In this case, the redundant pressure building module **300** brakes the vehicle alone.

[0059] The vehicle brake system **1** has a mode in which the master brake cylinder assembly **400** brakes alone. In this case, the master brake cylinder assembly **400** conveys the brake fluid to the wheel cylinder **100** alone. In this case, the master brake cylinder assembly **400** brakes the vehicle alone.

[0060] In some specific embodiments of the present disclosure, the vehicle brake system **1** has multiple braking modes that can be automatically switched. In different braking modes, at least two of the master pressure building module **200**, the redundant pressure building module **300**, and the master brake cylinder assembly **400** jointly convey the brake fluid to the wheel cylinder **100**.

[0061] For example, the vehicle brake system **1** has a mode in which the master pressure building module **200** and the redundant pressure building module **300** jointly build a pressure. In this case, two cases are divided. A first case is as follows. The master pressure building module **200** conveys the brake fluid to the wheel cylinders **100**. The redundant pressure building module **300** conveys the brake fluid to at least one wheel cylinder **100**. In this case, at least one wheel cylinder **100** receives the brake fluid of the master pressure building module **200** and the brake fluid of the redundant pressure building module **300**, and the braking force is higher. A second case is as follows. The master pressure building module **200** conveys the brake fluid to one part of the wheel cylinders **100**. The redundant pressure building module **300** conveys the brake fluid to another part of the wheel cylinders **100**. The brake fluid is conveyed more accurately.

[0062] The vehicle brake system **1** a mode in which the master pressure building module **200** and the master brake cylinder assembly **400** jointly build a pressure. In this case, two cases are divided. A first case is as follows. The master pressure building module **200** conveys the brake fluid to the wheel cylinders **100**. The master brake cylinder assembly **400** conveys the brake fluid to the wheel cylinders **100**. In this case, all the wheel cylinders **100** receive the brake fluid of the master pressure building module **200** and the brake fluid of the master brake cylinder assembly **400**, and the braking force is higher. A second case is as follows. The master pressure building module **200** conveys the brake fluid to one part of the wheel cylinders **100**. The master brake cylinder assembly **400** conveys the brake fluid to another part of the wheel cylinders **100**. The brake fluid is conveyed more accurately.

[0063] In some specific embodiments of the present disclosure, as shown in FIG. **1**, the master brake cylinder assembly **400** and the master pressure building module **200** are arranged separately and spaced away. In this way, the master brake cylinder assembly **400** and the master pressure building module **200** are not in direct contact. It may be understood that the vehicle brake system **1** frequently uses the master pressure building module **200** during braking. By spacing the master brake cylinder assembly **400** and the master pressure building module **200**, vibration generated during operation of the master pressure building module **200** is not transmitted to the master brake cylinder assembly **400**, which prevents the master brake cylinder assembly **400** from producing vibration and effectively avoids vibration noise, thereby improving riding comfort.

[0064] In addition, the master brake cylinder assembly **400** and the master pressure building module **200** are separately arranged, so that the master brake cylinder assembly **400** and the master pressure building module **200** may be separately arranged. In addition, a volume of each of the master brake cylinder assembly **400** and the master pressure building module **200** is relatively small. As a result, mounting and arrangement are more convenient, and layout is also more flexible.

[0065] In some specific embodiments of the present disclosure, as shown in FIG. 1, the master brake cylinder assembly **400** is mounted to the redundant pressure building module **300**. The volume of the redundant pressure building module **300** is smaller than that the master pressure building module **200**. By mounting the master brake cylinder assembly **400** and the redundant pressure building module **300** together, an entire volume of the master brake cylinder assembly **400** and the redundant pressure building module **300** are not excessively large, thereby facilitating arrangement. In addition, the redundant pressure building module **300** inputs the brake fluid to the wheel cylinder **100** only when the master pressure building module **200** is faulty or the braking force of the master pressure building module **200** is insufficient. In other words, the redundant pressure building module **300** has a relatively low operating frequency, and does not frequently transmit the vibration to the master brake cylinder assembly **400**. In addition, the master brake cylinder assembly **400** and the redundant pressure building module **300** can be disassembled together, which reduces disassembling steps and a required mounting space.

[0066] In some specific embodiments of the present disclosure, as shown in FIG. 1, the master pressure building module **200** is constructed with a first output flow channel **210**. The first output flow channel **210** is connected with the multiple wheel cylinders **100**. The redundant pressure building module **300** is constructed with a second output flow channel **310**. The second output flow channel **310** is connected with the at least one wheel cylinder **100**. The first output flow channel **210** may be a channel. The second output flow channel **310** may be a channel.

[0067] In other words, the master pressure building module **200** and the redundant pressure building module **300** are respectively in communication with the wheel cylinder **100** through different output flow channels. In this way, the master pressure building module **200** may convey the brake fluid to the multiple wheel cylinders **100** through the first output flow channel **210**. The redundant pressure building module **300** may convey the brake fluid to the at least one wheel cylinder **100** through the second output flow channel **310**. The conveying of the brake fluid to the multiple wheel cylinders **100** through the master pressure building module **200** does not interfere with the conveying of the brake fluid to the at least one wheel cylinder **100** through the redundant pressure building module **300**. In this way, the conveying of the brake fluid to the wheel cylinder **100** through the master pressure building module **200** is smoother, and the conveying of the brake fluid to the wheel cylinder **100** through the redundant pressure building module **300** is smoother, which help the master pressure building module **200** and the redundant pressure building module **300** respectively build a pressure in the wheel cylinder **100**, thereby improving the braking response speed of the vehicle brake system **1**.

[0068] In addition, when the first output flow channel **210** is damaged, a connection between the redundant pressure building module **300** and the wheel cylinder **100** is not affected at all. When the second output flow channel **310** is damaged, a connection between the master pressure building module **200** and the wheel cylinder **100** is not affected at all. The vehicle brake system **1** has higher redundancy, and the vehicle brake system **1** has more reliable braking.

[0069] In some specific embodiments of the present disclosure, as shown in FIG. 1, the vehicle brake system **1** further includes a first switching flow channel **410**.

[0070] The first switching flow channel **410** is respectively connected with the master pressure building module **200** and the master brake cylinder assembly **400**. The master brake cylinder assembly **400** is connected with the multiple wheel cylinders **100** through the first switching flow channel **410**, the master pressure building module **200**, and the first output flow channel **210**. The first switching flow channel **410** may be a channel or a pipe.

[0071] In this way, when the master pressure building module **200** and the redundant pressure building module **300** are faulty, the master brake cylinder assembly **400** may convey the brake fluid to the multiple wheel cylinders **100** through the first switching flow channel **410**.

[0072] In some specific embodiments of the present disclosure, as shown in FIG. **1**, the vehicle brake system **1** further includes a second switching flow channel **420**.

[0073] The second switching flow channel **420** is respectively connected with the redundant pressure building module **300** and the master brake cylinder assembly **400**. The master brake cylinder assembly **400** is connected with the multiple wheel cylinders **100** through the second switching flow channel **420**, the redundant pressure building module **300**, and the second output flow channel **310**. The second switching flow channel **420** may be a channel or a pipe.

[0074] In this way, when the master pressure building module **200** and the redundant pressure building module **300** are faulty, the master brake cylinder assembly **400** may convey the brake fluid to the multiple wheel cylinders **100** through the second switching flow channel **420**.

[0075] In some specific embodiments of the present disclosure, as shown in FIG. **1**, the vehicle brake system **1** may have both the first switching flow channel **410** and the second switching flow channel **420**, which helps increase a rate at which the master brake cylinder assembly **400** conveys the brake fluid to the multiple wheel cylinders **100**, thereby improving a braking response speed of the vehicle brake system **1**. In addition, the first switching flow channel **410** and the second switching flow channel **420** may have mutual redundancy, to improve braking reliability of the master brake cylinder assembly **400**.

[0076] In some specific embodiments of the present disclosure, as shown in FIG. **1**, the vehicle brake system **1** further includes a master control valve **220**, a redundant control valve **320**, and a master cylinder control valve **430**.

[0077] The master control valve **220** is arranged in the first output flow channel **210**, and is configured to control connection/disconnection between the master pressure building module **200** and the multiple wheel cylinders **100**. The redundant control valve **320** is arranged in the second output flow channel **310**, and is configured to control connection/disconnection between the redundant pressure building module **300** and the at least one wheel cylinder **100**. The master cylinder control valve **430** is mounted to the master pressure building module **200** and the redundant pressure building module **300**. The master cylinder control valve **430** is connected between the master brake cylinder assembly **400** and the multiple wheel cylinders **100**, and is configured to control connection/disconnection between the master brake cylinder assembly **400** and the multiple wheel cylinders **100**.

[0078] In this way, when the master pressure building module **200** can normally build a pressure and a pressure building capability can reach a required value, the master control valve **220** may be turned on, and the redundant control valve **320** and the master cylinder control valve **430** may be turned off. In this way, the master pressure building module **200** may convey the brake fluid to the multiple wheel cylinders **100**. The redundant pressure building module **300** and the master brake cylinder assembly **400** do not convey the brake fluid to the multiple wheel cylinders **100**. In this case, only the master pressure building module **200** can participate in braking.

[0079] When the master pressure building module **200** is faulty and the redundant pressure building module **300** can normally build a pressure, the redundant control valve **320** may be turned on, and the master control valve **220** and the master cylinder control valve **430** may be turned off. In this way, the redundant pressure building module **300** can convey the brake fluid to the at least one wheel cylinder **100**. The master pressure building module **200** and the master brake cylinder assembly **400** do not convey the brake fluid to the multiple wheel cylinders **100**. In this case, only the redundant pressure building module **300** can participate in braking.

[0080] When the pressure building capability of the master pressure building module **200** cannot reach the required value, the master control valve **220** and the redundant control valve **320** may be turned on, and the master cylinder control valve **430** may be turned off. In this way, the master

pressure building module **200** and the redundant pressure building module **300** may simultaneously convey the brake fluid to the multiple wheel cylinders **100**. The master brake cylinder assembly **400** does not convey the brake fluid to the multiple wheel cylinders **100**. The master pressure building module **200** and the redundant pressure building module **300** simultaneously participate in the braking, to ensure that the vehicle brake system **1** has a sufficient braking force and a fast braking response.

[0081] When the master pressure building module **200** and the redundant pressure building module **300** are both faulty, the master cylinder control valve **430** may be turned on, and the master control valve **220** and the redundant control valve **320** may be turned off. In this way, the master brake cylinder assembly **400** may directly convey the brake fluid to the multiple wheel cylinders **100**. The master pressure building module **200** and the redundant pressure building module **300** do not convey the brake fluid to the multiple wheel cylinders **100**. In this case, only the master brake cylinder assembly **400** can participate in the braking.

[0082] Therefore, a communication situation between the master pressure building module **200**, the redundant pressure building module **300**, and the master brake cylinder assembly **400** and the multiple wheel cylinders **100** may be adjusted through cooperative operation of the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430**, which has a simple structure and facilitates control.

[0083] In some specific embodiments of the present disclosure, as shown in FIG. **1** and FIG. **2**, the vehicle brake system **1** further includes a brake pedal **500**, a displacement sensor **510**, a vehicle wheel speed sensor **901**, a wheel cylinder pressure sensor **903**, and a controller **900**.

[0084] The brake pedal **500** is connected with the master brake cylinder assembly **400**. The displacement sensor **510** is configured to detect a movement position of at least one of the brake pedal **500** and a piston **450** of the master brake cylinder assembly **400**. The wheel cylinder pressure sensor **903** is connected with the multiple wheel cylinders **100**, and is configured to detect pressures of the multiple wheel cylinders **100**. The controller **900** is respectively connected with the displacement sensor **510**, the vehicle wheel speed sensor **901**, the wheel cylinder pressure sensor **903**, the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430**. The controller **900** is configured to control closing/opening of the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430** based on a detection result of the displacement sensor **510** and a detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**.

[0085] The displacement sensor **510** may detect only the movement position of the brake pedal **500**. Alternatively, the displacement sensor **510** may detect only the movement position of the piston **450** of the master brake cylinder assembly **400**. Alternatively, two displacement sensors **510** are provided. The two displacement sensors **510** respectively detect the movement position of the brake pedal **500** and the movement position of the piston **450** of the master brake cylinder assembly **400**.

[0086] Specifically, if the controller **900** detects a movement in the brake pedal **500** or the piston **450** of the master brake cylinder assembly **400**, it indicates that the driver has a braking intention and needs to brake the vehicle. The master pressure building module **200** starts to build a pressure and brakes. In addition, the controller **900** may obtain a vehicle speed through the vehicle wheel speed sensor **901**, to determine whether the vehicle is decelerating, and to determine whether the braking force outputted by relying on only the master pressure building module **200** can meet a braking force required by a current vehicle speed. In addition, the controller **900** may obtain pressures in the multiple wheel cylinders **100** based on a third controller, to determine whether the master pressure building module **200** or the redundant pressure building module **300** can normally build a pressure.

[0087] The controller **900** may simultaneously obtain detection information of the displacement sensor **510**, detection information of the vehicle wheel speed sensor **901**, and detection information

of the wheel cylinder pressure sensor **903**. Therefore, when a driver steps on the brake pedal **500**, the controller **900** may synchronously obtain the vehicle speed and the pressures in the multiple wheel cylinders **100** when the controller **900** obtains the movement position of the brake pedal **500** or the movement position of the piston **450** of the master brake cylinder assembly **400**. Therefore, the controller **900** has a higher response speed.

[0088] Alternatively, the controller **900** may first obtain the detection information of the displacement sensor **510**. If the controller **900** detects a movement of the brake pedal **500** or the master brake cylinder assembly **400**, the controller **900** is configured to: obtain the detection information of the vehicle wheel speed sensor **901** and the detection information of the wheel cylinder pressure sensor **903** again; and control closing/opening of the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430** based on a detection result of the displacement sensor **510** and a detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**. If the controller **900** does not detect a movement of the brake pedal **500** or the master brake cylinder assembly **400**, the controller does not need to obtain the detection information of the vehicle wheel speed sensor **901** and the detection information of the wheel cylinder pressure sensor **903**. In this way, operation steps of the controller **900** may be simplified, which help reduce costs of the controller **900**.

[0089] In some specific embodiments of the present disclosure, the controller **900** is further respectively connected with the master pressure building module **200** and the redundant pressure building module **300**. The controller **900** is configured to control, based on the detection result of the displacement sensor **510** and the detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**, the master pressure building module **200** and the redundant pressure building module **300** whether to build a pressure.

[0090] In this way, if the controller **900** detects a movement of the brake pedal **500** or the piston **450** of the master brake cylinder assembly **400**s, and the controller **900** determines that the pressure building capability of the master pressure building module **200** meets a demand at the current vehicle speed, the controller **900** controls the master pressure building module **200** to build a pressure. If the vehicle does not decelerate or the pressures in the multiple wheel cylinders **100** do not change after the master pressure building module **200** builds the pressure, it indicates that the master pressure building module **200** fails. In this case, the master control valve **220** may be turned off and the redundant control valve **320** may be turned on. In other words, the controller **900** controls the master pressure building module **200** to stop building the pressure and controls the redundant pressure building module **300** to build a pressure. If the vehicle still does not decelerate or the pressures in the wheel cylinders **100** still do not change, it indicates that the redundant pressure building module **300** fails. In this case, the redundant control valve **320** and the master control valve **220** may be turned off. The controller **900** controls both the master pressure building module **200** and the redundant pressure building module **300** not to build the pressure, and turns on the master cylinder control valve **430**, to directly convey the brake fluid to the multiple wheel cylinders **100** through the master brake cylinder assembly **400**, so as to brake the vehicle.

[0091] In this way, the controller **900** is configured to control, based on the detection result of the displacement sensor **510** and the detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**, the master pressure building module **200** and the redundant pressure building module **300** whether to build a pressure, so that the vehicle brake system **1** can quickly perform the braking response and meet the braking demand of the vehicle. In addition, when the master pressure building module **200** is faulty, the controller **900** may control the master pressure building module **200** to stop building the pressure, to avoid further failure and damage of the master pressure building module **200**. In addition, when the redundant pressure building module **300** is faulty, the controller **900** may control the redundant pressure building module **300** to stop building the pressure, to avoid further failure and damage of the redundant pressure building module **300**, which helps improve reliability of the vehicle brake system **1**.

[0092] In some specific embodiments of the present disclosure, as shown in FIG. 1 to FIG. 3, multiple controllers **900** are provided, which include a master module controller **9001** and a redundant module controller **9002**. The master module controller **9001** is respectively connected with the displacement sensor **510**, the vehicle wheel speed sensor **901**, the wheel cylinder pressure sensor **903**, the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430**. The master module controller **9001** is configured to control, based on the detection result of the displacement sensor **510** and the detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**, closing/opening of the master control valve **220** and the master pressure building module **200** whether to build a pressure.

[0093] The redundant module controller **9002** is respectively connected with the displacement sensor **510**, the vehicle wheel speed sensor **901**, the wheel cylinder pressure sensor **903**, the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430**. The redundant module controller **9002** is configured to control, based on the detection result of the displacement sensor **510** and the detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**, closing/opening of the redundant control valve **320** and whether the redundant pressure building module **300** to build a pressure.

[0094] The master module controller **9001** or the redundant module controller **9002** is connected with the displacement sensor **510**, the vehicle wheel speed sensor **901**, the wheel cylinder pressure sensor **903**, the master control valve **220**, the redundant control valve **320**, and the master cylinder control valve **430**. The master module controller **9001** is configured to control, based on the detection result of the displacement sensor **510** and the detection result of at least one of the vehicle wheel speed sensor **901** and the wheel cylinder pressure sensor **903**, the closing/opening of the master cylinder control valve **430**.

[0095] In some specific embodiments of the present disclosure, as shown in FIG. 1, the vehicle brake system **1** further includes a first angle sensor **230**, a first pressure sensor **240**, a second angle sensor **330**, a second pressure sensor **340**, and a third pressure sensor **440**.

[0096] The first angle sensor **230** is configured to detect an angle change of a motor of the master pressure building module **200**. The first pressure sensor **240** is configured to detect a pressure of a conveyed brake fluid of the master pressure building module **200**. The second angle sensor **330** is configured to detect an angle change of a motor of the redundant pressure building module **300**. The second pressure sensor **340** is configured to detect a pressure of a conveyed brake fluid of the redundant pressure building module **300**. The third pressure sensor **440** is configured to detect a pressure of a conveyed brake fluid of the master brake cylinder assembly **400**.

[0097] The controller **900** is further respectively with the first angle sensor **230**, the first pressure sensor **240**, the second angle sensor **330**, the second pressure sensor **340**, and the third pressure sensor **440**. The controller **900** is configured to: determine whether the master pressure building module **200** is faulty based on detection results of the first angle sensor **230** and the first pressure sensor **240**, determine whether the master pressure building module **200** is faulty based on detection results of the second angle sensor **330** and the second pressure sensor **340**, and determine whether the master brake cylinder assembly **400** is faulty based on a detection result of the third pressure sensor **440**.

[0098] Specifically, if the controller **900** controls the master pressure building module **200** to build a pressure, but the controller **900** detects an angle change of the motor of the master pressure building module **200** through the first angle sensor **230**, it indicates that the motor of the master pressure building module **200** is faulty. If the controller **900** detects an abnormal change in the pressure of the brake fluid conveyed by the master pressure building module **200** through the first pressure sensor **240**, it indicates that the master pressure building module **200** cannot normally convey the brake fluid.

[0099] In addition, if the controller **900** controls the redundant pressure building module **300** to build a pressure, but the controller **900** does not detect a change in the angle of the motor of the

redundant pressure building module **300** through the second angle sensor **330**, it indicates that the motor of the redundant pressure building module **300** is faulty. If the controller **900** detects an abnormal change in the pressure of the brake fluid conveyed by the redundant pressure building module **300** through the second pressure sensor **340**, it indicates that the redundant pressure building module **300** cannot normally convey the brake fluid.

[0100] In addition, if the controller **900** controls both the master pressure building module **200** and the redundant pressure building module **300** to stop building the pressure, and the master brake cylinder assembly **400** directly conveys the brake fluid to the multiple wheel cylinders **100**, the controller **900** obtains detection information of the third pressure sensor **440**, and may determine whether the pressure of the brake fluid conveyed by the master brake cylinder assembly **400** is abnormal based on the detection information of the wheel cylinder pressure sensor **903**, so as to determine whether the master brake cylinder assembly **400** is faulty.

[0101] In some specific embodiments of the present disclosure, as shown in FIG. 1, the vehicle brake system **1** further includes a master liquid reservoir **250** and a redundant liquid reservoir **350**.

[0102] The master liquid reservoir **250** is connected with the master pressure building module **200**. The redundant liquid reservoir **350** is respectively connected with the redundant pressure building module **300**, the master brake cylinder assembly **400**, and the master liquid reservoir **250**.

[0103] In this way, an external pipeline does not need to be additionally arranged. The master liquid reservoir **250** may directly provide the master pressure building module **200** with the brake fluid. The redundant liquid reservoir **350** may directly provide the redundant pressure building module **300** with the brake fluid. Therefore, the brake fluid can quickly reach the master pressure building module **200** or the redundant pressure building module **300**, a pressure building speed is higher, and a braking response of the vehicle brake system **1** is faster.

[0104] In addition, the redundant liquid reservoir **350** is connected with the master liquid reservoir **250**. If a relatively small amount of brake fluid exists in the redundant liquid reservoir **350**, the master liquid reservoir **250** may supplement the redundant liquid reservoir **350** with the brake fluid. Similarly, if a relatively small amount of brake fluid exists in the master liquid reservoir **250**, the redundant liquid reservoir **350** may supplement the master liquid reservoir **250** with the brake fluid, so as to ensure that the brake fluid in the master liquid reservoir **250** and the brake fluid in the redundant liquid reservoir **350** are both relatively sufficient, to meet a pressure building demand of the master pressure building module **200** and a pressure building demand of the redundant pressure building module **300**.

[0105] In some specific embodiments of the present disclosure, as shown in FIG. 1, the vehicle brake system **1** further includes a brake pedal **500**, a stroke simulator **600**, and a simulator switch valve **610**.

[0106] The brake pedal **500** is connected with the master brake cylinder assembly **400**. The stroke simulator **600** is respectively connected with the master brake cylinder assembly **400** and the brake pedal **500**, and is configured to feed back a damping force to the brake pedal **500**. The simulator switch valve **610** is respectively connected with the stroke simulator **600** and the master brake cylinder assembly **400**, and is configured to control on/off of the stroke simulator **600** and the master brake cylinder assembly **400**.

[0107] Because the stroke simulator **600** is arranged, when the driver steps on the brake pedal **500**, the stroke simulator **600** may feed back a damping force to the brake pedal **500**, to indirectly feed back the damping force to a foot of the driver, and the driver may determine, based on the damping force that is fed back, whether to step on and move the brake pedal **500**. In this way, the driver can slowly step on the brake pedal **500**, to avoid an excessively sensitive braking response as a result of the driver suddenly stepping on the brake pedal **500** by an excessively large distance. Therefore, more comfortable vehicle brake is achieved.

[0108] In some specific embodiments of the present disclosure, as shown in FIG. 1, the vehicle brake system **1** further includes multiple pressure increasing valves **700** and multiple pressure

reducing valves **800**.

[0109] The multiple pressure increasing valves **700** are connected with the multiple wheel cylinders **100** in one-to-one correspondence. Each pressure increasing valve **700** is respectively connected with the master pressure building module **200**, the redundant pressure building module **300**, and the master brake cylinder assembly **400**. The multiple pressure reducing valves **800** are connected with the multiple wheel cylinders **100** in one-to-one correspondence. Each pressure reducing valve **800** is connected with the liquid reservoir.

[0110] In this way, when the brake pedal **500** is stepped, the controller **900** controls the multiple pressure increasing valves **700** to be turned on and the pressure reducing valves **800** to be turned off, so that the brake fluid can enter the multiple wheel cylinders **100**. When a wheel cylinder **100** has a sufficient pressure, in other words, when the pressure of the wheel cylinder **100** meets the braking demand, the pressure increasing valve **700** corresponding to the wheel cylinder **100** may be turned off. In addition, because the multiple pressure increasing valves **700** are in one-to-one correspondence with the multiple wheel cylinders **100**, each pressure increasing valve **700** can control the wheel cylinder **100** corresponding thereto, so that more accurate pressure control of the wheel cylinder **100** is achieved.

[0111] When the brake pedal **500** is reset, the multiple pressure reducing valves **800** may be turned on and the multiple pressure increasing valves **700** may be turned off, to guide the brake fluid of the multiple wheel cylinders **100** to the liquid reservoir (for example, the master liquid reservoir **250** or the redundant liquid reservoir **350**), so as to reduce the pressures of the wheel cylinders **100**. The brake fluid may be reused, to reduce costs. Moreover, because the multiple pressure reducing valves **800** are in one-to-one correspondence with the multiple wheel cylinders **100**, each pressure reducing valve **800** can reduce the pressure of the wheel cylinder **100** corresponding thereto, so that more accurate pressure control of the wheel cylinder **100** is achieved.

[0112] In other words, through the coordination between the multiple pressure increasing valves **700** and the multiple pressure reducing valves **800**, an amount of the brake fluid in each wheel cylinder **100** can be adjusted, to adjust the pressure of each wheel cylinder **100**.

[0113] A vehicle **1000** in an embodiment of the present disclosure is described below with reference to FIG. 4. The vehicle **1000** includes the vehicle brake system **1** in the foregoing embodiments of the present disclosure.

[0114] The vehicle in the embodiment of the present disclosure uses the vehicle brake system **1** in the foregoing embodiments of the present disclosure, which has advantages such as strong braking force output, a high braking response speed, and high braking safety.

[0115] Other compositions and operations of the vehicle brake system **1** and the vehicle having the same in the embodiments of the present disclosure are known to a person of ordinary skill in the art. Details are not described herein again.

[0116] In the descriptions of this specification, descriptions using reference terms “an embodiment”, “some embodiments”, “an exemplary embodiment”, “an example”, “a specific example”, or “some examples” mean that specific characteristics, structures, materials, or features described with reference to the embodiment or example are included in at least one embodiment or example of the present disclosure. In this specification, exemplary descriptions of the foregoing terms do not necessarily refer to the same embodiment or example.

[0117] Although the embodiments of the present disclosure have been shown and described, a person of ordinary skill in the art should understand that various changes, modifications, replacements, and variations may be made to the embodiments without departing from the principles and spirit of the present disclosure, and the scope of the present disclosure is as defined by the appended claims and their equivalents.

Claims

- 1.** A vehicle brake system, comprising: a plurality of wheel cylinders; a master pressure building module, respectively connected with the plurality of wheel cylinders, and configured to convey a brake fluid to the plurality of wheel cylinders; a redundant pressure building module, connected with at least one wheel cylinder, and configured to convey a brake fluid to the at least one wheel cylinder; and a master brake cylinder assembly, respectively connected with the plurality of wheel cylinders, and configured to convey a brake fluid to the plurality of wheel cylinders.
- 2.** The vehicle brake system according to claim 1, further comprising a plurality of braking modes between which the vehicle brake system is automatically switchable, wherein, in different braking modes, the master pressure building module, the redundant pressure building module, or the master brake cylinder assembly separately conveys a brake fluid to the wheel cylinders, or the master pressure building module and the redundant pressure building module jointly convey a brake fluid to the wheel cylinders, or the master pressure building module and the master brake cylinder assembly jointly convey a brake fluid to the wheel cylinders, or the redundant pressure building module and the master brake cylinder assembly jointly convey a brake fluid to the wheel cylinders.
- 3.** The vehicle brake system according to claim 1, wherein the master brake cylinder assembly and the master pressure building module are arranged separately and spaced away.
- 4.** The vehicle brake system according to claim 2, wherein the master brake cylinder assembly and the master pressure building module are arranged separately and spaced away.
- 5.** The vehicle brake system according to claim 1, wherein the master brake cylinder assembly is mounted to the redundant pressure building module.
- 6.** The vehicle brake system according to claim 2, wherein the master brake cylinder assembly is mounted to the redundant pressure building module.
- 7.** The vehicle brake system according to claim 3, wherein the master brake cylinder assembly is mounted to the redundant pressure building module.
- 8.** The vehicle brake system according to claim 1, wherein a first output flow channel is constructed in the master pressure building module; the first output flow channel is connected with the plurality of wheel cylinders; a second output flow channel is constructed in the redundant pressure building module; and the second output flow channel is connected with the at least one wheel cylinder.
- 9.** The vehicle brake system according to claim 7, wherein a first output flow channel is constructed in the master pressure building module; the first output flow channel is connected with the plurality of wheel cylinders; a second output flow channel is constructed in the redundant pressure building module; and the second output flow channel is connected with the at least one wheel cylinder.
- 10.** The vehicle brake system according to claim 8, further comprising: a first switching flow channel, respectively connected with the first output flow channel and the master brake cylinder assembly, wherein the master brake cylinder assembly is connected with the plurality of wheel cylinders through the first switching flow channel.
- 11.** The vehicle brake system according to claim 1, further comprising: a second switching flow channel, respectively connected with the second output flow channel and the master brake cylinder assembly, wherein the master brake cylinder assembly is connected with the plurality of wheel cylinders through the second switching flow channel.
- 12.** The vehicle brake system according to claim 8, further comprising: a master control valve, arranged in the first output flow channel, and configured to control connection/disconnection between the master pressure building module and the plurality of wheel cylinders; a redundant control valve, arranged in the second output flow channel, and configured to control connection/disconnection between the redundant pressure building module and the at least one wheel cylinder; and a master cylinder control valve, mounted to at least one of the master pressure building module and the redundant pressure building module, wherein the master cylinder control valve is connected between the master brake cylinder assembly and the plurality of wheel cylinders, and is configured to control connection/disconnection between the master brake cylinder

assembly and the plurality of wheel cylinders.

13. The vehicle brake system according to claim 12, further comprising: a brake pedal, connected with the master brake cylinder assembly; a displacement sensor, configured to detect a movement position of at least one of the brake pedal and a piston of the master brake cylinder assembly; a wheel cylinder pressure sensor, connected with the plurality of wheel cylinders, and configured to detect pressures of the plurality of wheel cylinders; and a controller, respectively connected with the displacement sensor, a vehicle wheel speed sensor, the wheel cylinder pressure sensor, the master control valve, the redundant control valve, and the master cylinder control valve, wherein the controller is configured to control closing/opening of the master control valve, the redundant control valve, and the master cylinder control valve based on a detection result of the displacement sensor and a detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor.

14. The vehicle brake system according to claim 1, wherein the controller is further respectively connected with the master pressure building module and the redundant pressure building module; and the controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, the master pressure building module and the redundant pressure building module whether to build a pressure.

15. The vehicle brake system according to claim 1, wherein a plurality of controllers are arranged, the plurality of controllers comprise a master module controller and a redundant module controller; the master module controller is respectively connected with the displacement sensor, the vehicle wheel speed sensor, the wheel cylinder pressure sensor, the master control valve, the redundant control valve, and the master cylinder control valve; the master module controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, closing/opening of the master control valve and the master pressure building module whether to build a pressure; the redundant module controller is respectively connected with the displacement sensor, the vehicle wheel speed sensor, the wheel cylinder pressure sensor, the master control valve, the redundant control valve, and the master cylinder control valve; and the master module controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, closing/opening of the redundant control valve and whether the redundant pressure building module to build a pressure, wherein the master module controller or the redundant module controller is configured to control, based on the detection result of the displacement sensor and the detection result of at least one of the vehicle wheel speed sensor and the wheel cylinder pressure sensor, the closing/opening of the master cylinder control valve.

16. The vehicle brake system according to claim 13, further comprising: a first angle sensor, configured to detect an angle change of a motor of the master pressure building module; a first pressure sensor, configured to detect a pressure of a brake fluid conveyed by the master pressure building module; a second angle sensor, configured to detect an angle change of a motor of the redundant pressure building module; a second pressure sensor, configured to detect a pressure of a brake fluid conveyed by the redundant pressure building module; and a third pressure sensor, configured to detect a pressure of a brake fluid conveyed by the master brake cylinder assembly, wherein the controller is further respectively with the first angle sensor the first pressure sensor, the second angle sensor, the second pressure sensor, and the third pressure sensor; and the controller is configured to: determine whether the master pressure building module is faulty based on detection results of the first angle sensor and the first pressure sensor, determine whether the redundant pressure building module is faulty based on detection results of the second angle sensor and the second pressure sensor, and determine whether the master brake cylinder assembly is faulty based on a detection result of the third pressure sensor.

- 17.** The vehicle brake system according to claim 1, further comprising: a master liquid reservoir, connected with the master pressure building module; and a redundant liquid reservoir, respectively connected with the redundant pressure building module, the master brake cylinder assembly, and the master liquid reservoir.
- 18.** The vehicle brake system according to claim 1, further comprising: a brake pedal, connected with the master brake cylinder assembly; a stroke simulator, respectively connected with the master brake cylinder assembly and the brake pedal, and configured to feed back a damping force to the brake pedal; and a simulator switch valve, respectively connected with the stroke simulator and the master brake cylinder assembly, and configured to control on/off the stroke simulator and the master brake cylinder assembly.
- 19.** The vehicle brake system according to claim 1, further comprising: a plurality of pressure increasing valves, connected with the plurality of wheel cylinders in one-to-one correspondence, wherein each pressure increasing valve is respectively connected with the master pressure building module, the redundant pressure building module, and the master brake cylinder assembly; and a plurality of pressure reducing valves, connected with the plurality of wheel cylinders in one-to-one correspondence, wherein each pressure reducing valve is connected with the liquid reservoir.
- 20.** A vehicle, comprising the vehicle brake system according to claim 1.
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